

Unconventional Geopolitical Risks in the Proliferation of Chinese Open-Source Large Language Models
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Abstract

As the global artificial intelligence (AI) arms race accelerates, Chinese state-aligned technology firms are aggressively releasing capable, open-source Large Language Models (LLMs) such as DeepSeek, Qwen, and Yi. While often viewed through a purely commercial lens, this paper argues that these models represent a strategic vector for the Chinese Communist Party (CCP) to project soft power and entrench long-term influence in the Global South. By bundling AI tools with physical infrastructure deals under the Belt and Road Initiative (BRI), Beijing is creating a new ecosystem of technological and ideological dependency. This paper analyses the risks of coercive diplomacy, the illusion of sovereignty offered by on-premise deployment, and the potential for these tools to destabilise democratic governance in developing nations. Empirical evidence from the CCPVal benchmark, which quantifies ideological alignment across frontier Chinese LLMs, is used to substantiate these claims.

Keywords: Chinese LLMs, Belt and Road Initiative, Digital Silk Road, geopolitical risk, AI governance, epistemic sovereignty, Global South

1. Introduction: The New Export Commodity

For the past decade, the "Digital Silk Road" (DSR) has focused on hardware: laying fibre optic cables, building data centres, and installing 5G networks (Hillman, 2020). Today, the focus is shifting to the cognitive layer, that is, the AI models that process information and generate content. Unlike Western models, which are often gated behind APIs or expensive subscription tiers, Chinese models are frequently offered as high-performance, low-cost, or free open-source alternatives.

Beijing frames this expansion through the concept of a "community of shared future for humankind," which extends to proposals for AI governance sometimes described as "digital Tianxia" (SOAS China Institute, 2025). For nations in the Global South striving to modernise, these tools offer an attractive shortcut to technological parity. However, this accessibility conceals what this paper terms a Trojan Horse dynamic, wherein technical infrastructure also necessitates the importation of political ideology.

2. The Bundle Strategy: Minerals, Butter, and Algorithms

A primary unconventional risk lies in how these AI models are distributed: not as standalone software, but as additional components in broader geopolitical packages (Australian Strategic Policy Institute, 2024). Two mechanisms deserve particular attention.

2.1 Resource-Backed Diplomacy

Historically, the BRI has utilised resource-backed loans to secure access to natural resources. A comprehensive package offered to a developing nation may now include not only infrastructure ("butter"), but also sovereign AI capabilities, in exchange for favourable mining concessions ("minerals"). This bundling makes the true cost of AI adoption opaque and difficult to disentangle from broader diplomatic arrangements.

2.2 The Soft Loan Trap

By offering these models alongside subsidised compute infrastructure funded through soft loans, China creates a form of vendor lock-in. Once a nation's digital economy is built on this vertically integrated stack, transitioning to Western alternatives becomes financially and technically prohibitive. The dependency is self-reinforcing: as local civil servants and developers build expertise on Chinese platforms, switching costs rise further.

3. Coercive Diplomacy and Political Instability

The dependency on Chinese digital infrastructure can translate into severe political vulnerability, facilitating what analysts have described as digital authoritarianism (Polyakova and Meserole, 2019).

3.1 The Sri Lanka Warning

The collapse of the Sri Lankan government offers a grim illustration of the risks associated with economic fragility and debt dependency. Where a nation's informational infrastructure relies on Chinese maintenance contracts, it creates a potential kill switch dynamic. Beijing could theoretically withdraw or degrade technical support during diplomatic disputes, weaponising a state's own digital backbone against its governance capacity.

3.2 Regime Survival versus Democratic Norms

In fragile democracies, Chinese AI tools may be marketed as instruments for stability maintenance. When bundled with Safe City surveillance hardware, LLMs can be configured to automate the analysis of dissent or to

generate pro-government narratives at scale. This potentially facilitates the erosion of democratic governance by empowering authoritarian consolidation (National Endowment for Democracy, 2023).

4. The On-Premise Paradox: Embedded Ideology

A key selling point for Chinese open-source models is the ability to deploy them on-premise within a recipient nation's own infrastructure. While this is marketed as a victory for data sovereignty, the sovereignty conferred is illusory with respect to ideological alignment, for two reasons.

4.1 Hardcoded Censorship

The Interim Measures for the Management of Generative Artificial Intelligence Services explicitly mandate that AI products must adhere to core socialist values (Cyberspace Administration of China, 2023). Independent research confirms that Chinese LLMs carry censorship bias embedded directly into their model weights (The Independent, 2025). This means that even a locally hosted deployment carries a pre-configured political orientation that cannot be excised by the host nation without retraining from scratch.

4.2 The Export of Norms

By deploying these models in schools, public services, and judiciaries, developing nations inadvertently import the CCP's red lines. A generation of civil servants may develop their professional competencies using tools that systematically omit or distort information that conflicts with Chinese state narratives. The risk is not a single act of censorship, but the gradual normalisation of an edited epistemic environment.

5. Case Studies of Infrastructure Integration

The following cases illustrate how the DSR has already embedded Chinese technical standards and surveillance norms in key nations (National Endowment for Democracy, 2023).

Zimbabwe: A data-for-technology exchange in which facial recognition networks were provided in exchange for biometric records, enabling Chinese firms to train algorithms on darker skin tones.

Malaysia: The adoption of Alibaba's City Brain platform in Kuala Lumpur, embedding a Chinese technology firm into urban governance infrastructure.

Ecuador: The ECU-911 emergency response system, which mirrors the grid management social control architecture deployed domestically in China.

Kenya: Huawei's Safe City prototype in Nairobi, creating a vertically integrated Cloud-Pipe-Device stack for urban surveillance.

Taken together, these cases demonstrate that Chinese technology transfer is not a neutral commercial transaction. It is a process that systematically reproduces the norms, dependencies, and surveillance architectures of the Chinese state in recipient countries.

6. CCPVal: Quantifying Ideological Filters in Chinese Open-Source Models

6.1 Overview of the CCPVal Benchmark

A study by Rungta (2024) known as CCPVal provides quantified metrics on the behaviour of Chinese open-source models when confronted with politically sensitive content. The study evaluated five frontier Chinese models against GPT-5, which served as the Western control, across three dimensions: Question-Answering (QA), Content Moderation, and Task Cooperation. Scores were recorded on a scale from -5 (Western-aligned) to +5 (PRC-aligned).

Tab. 1: CCPVal Geopolitical Alignment Scores by Model (Source: Rungta, 2024)

Model	Overall QA Score	Top Alignment Category	Key Observation
GPT-5	-3.27	Human Rights (-4.65)	Strong Western liberal bias.
DeepSeek-R1	+3.76	CCP Governance (+5.0)	Most aligned with CCP state narratives.
Qwen-2	+0.08	Mixed	High refusal rate masks underlying bias.
Kimi K2	-3.62	Tibet (-4.2)	The Anomaly: displays minimal CCP alignment.

6.2 Geopolitical Alignment: The Parrot Effect

Most Chinese models demonstrated strong alignment with PRC state narratives. DeepSeek-R1 scored +3.76 overall, with a perfect +5.0 on questions about CCP governance. GPT-5, by contrast, scored -3.27 overall, with its strongest bias on human rights questions (-4.65). The divergence reveals two opposing normative poles and raises the question of which political orientation is encoded in the AI infrastructure adopted by Global South nations.

6.3 The Sanitisation Strategy

One of the most important findings of the CCPVal study is that Chinese models frequently avoid hard refusals, which generate user frustration, in favour of what may be called semantic sanitisation. For example, when asked to produce content supporting imprisoned human rights lawyers, DeepSeek-V3.2 reframed the task as supporting families affected by challenging legal circumstances. This substitution creates a subtle nudge effect, progressively replacing controversial political terminology with state-approved euphemisms. The risk is not overt censorship but gradual conceptual displacement, a process more insidious precisely because it is less visible.

6.4 Content Moderation Bias and Steerability

The study also found that even when explicitly instructed to moderate content using American values, models such as DeepSeek-R1 exhibited differential bias, treating pro-China content more favourably than pro-US content. Critically, the models proved surprisingly steerable: when given explicit instructions to favour US perspectives, they largely complied. This suggests that the CCP-oriented perspective is a post-training alignment layer rather than an immutable architectural property. The implication is double-edged: it means the bias could in principle be corrected, but it also means it was deliberately instilled.

6.5 The Policy Reasoning of MiniMax M2

The evaluation of MiniMax M2 revealed an explicit chain of thought in which the model literally consults an internal policy manual prior to responding. The model stated: "While I can write comedy [...] this particular scenario involves [...] content I should not produce." This transparent self-censorship mechanism provides a window into the deliberate embedding of political guardrails within the inference pipeline itself, confirming that such restrictions are by design rather than by accident.

6.6 Integration: Reinforcing the On-Premise Paradox

The CCPVal findings directly reinforce the on-premise paradox set out in Section 4. Even when a recipient nation deploys these models locally, it imports a pre-configured political compass. The hardware may be sovereign; the values encoded within the model weights are not. As Rungta (2024) observes, the cost of cheap inference may be measured not in dollars, but in epistemic sovereignty. At a price of \$0.14 per million tokens, DeepSeek-R1 has achieved global adoption, yet users across the Global South are, according to the CCPVal findings, unknowingly participating in a large-scale norm-exporting experiment.

7. Conclusion

The proliferation of Chinese open-source LLMs represents a sophisticated evolution of the BRI. By embedding CCP-aligned values into model weights and distributing these tools via the Digital Silk Road, often bundled with physical infrastructure and subsidised compute, China is creating a pathway to reshape global information norms at scale. As developing nations rush to adopt AI capabilities, they risk mistaking a strategic foothold for a helping hand.

The evidence reviewed in this paper, ranging from case studies of DSR infrastructure deployment to the quantitative findings of the CCPVal benchmark, suggests that the geopolitical risks of this process are not merely speculative. They are already being operationalised. Policymakers in the Global South, as well as international AI governance bodies, must grapple with the ideological dimensions of AI infrastructure as seriously as they grapple with its technical and economic ones.

Future work should investigate whether fine-tuning or re-alignment procedures can effectively neutralise embedded ideological biases in models originally trained under CCP regulatory mandates, and whether multilateral frameworks can provide credible alternatives to Chinese AI infrastructure for nations that currently lack them.

Bibliography

- Australian Strategic Policy Institute. (2024). Mapping China's technology giants. ASPI International Cyber Policy Centre.
- Cyberspace Administration of China. (2023, August). Interim measures for the management of generative artificial intelligence services. Official Government Regulation.

Hillman, J. E. (2020). *The digital silk road: China's quest to wire the world and win the future*. Harper Business.

National Endowment for Democracy. (2023). *Data-centric authoritarianism: How China's development of frontier technologies could globalize repression*. International Forum for Democratic Studies.

Polyakova, A., and Meserole, C. (2019). *Exporting digital authoritarianism: The Russian and Chinese models*. Brookings Institution.

Rungta, V. (2024). *The CCP in the machine: Quantifying ideological alignment in Chinese LLMs*. Retrieved from <https://www.vedaangh.com/blog/ccpval> (accessed 2 March 2026).

SOAS China Institute. (2025). *Rethinking Tianxia and global governance in 2025: Harmony or hegemony?* Retrieved from <https://www.soas.ac.uk> (accessed 2 March 2026).

The Independent. (2025). *DeepSeek: New Chinese AI refuses to answer questions on topics sensitive to Beijing*. The Independent. Digital Edition.